GAO YUNZHE

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EDUCATION

Nanyang Technological University (NTU) Singapore
M.Sc. in Computer Control and Automation **Aug. 2025 – Jan. 2027 (Expected)**

- **GPA: 4.63 / 5.0**
- Honors: EEE Global Merit Award (Top 2% of the cohort; ∼20 recipients out of 1,000+ eligible students).
- Advisor: Prof. Mao Kezhi
- Dissertation Topic: Design and Deployment of Lightweight Large Language Models (LLM) for Efficient, Localized Deployment.
- Selected Coursework: Artificial Intelligence & Data Mining, Genetic Algorithms & Machine Learning, Computer Control Networks.

Northeastern University (NEU) Shenyang, China
B.E. in Automation **Sept. 2021 – Jul. 2025**

- **GPA: 3.76 / 5.0 (Avg. Score: 87.6/100)**
- Honors: University Academic Scholarship.
- Selected Coursework: Artificial Intelligence Foundation (93/100); Distributed Cooperative Control (96/100); Cloud Computing and Industrial Internet (87/100).

RESEARCH EXPERIENCE

Energy-Efficient Edge-Cloud Collaborative LLM Inference NTU, Singapore
Master's Dissertation (Advisor: Prof. Mao Kezhi) Aug. 2025 – Present

- Developed Green-Edge-Cloud, a collaborative LLM inference framework that uses the student model's length-normalized sequence likelihood to select teacher calls without an auxiliary routing network.
- Distilled Qwen2.5-72B reasoning into a 1.5B student with chain-of-thought (CoT) supervision and LoRA on 6,340 examples; improved confidence AUROC on GSM8K from 0.6544 to 0.6996.
- Calibrated thresholds on 400 held-out questions and froze them before testing. Achieved 81.27% accuracy with 24.34% teacher calls on GSM8K; transferring the same threshold to SVAMP yielded 85.67% accuracy with 34.67% teacher calls.
- Built vLLM services and GPU power sampling across three GPUs with both models resident, integrating energy over complete requests, including student generation, teacher processing, and waiting periods.
- In a separate 150-question online evaluation repeated three times, reduced GPU energy by 56.42% and mean response latency from 27.99 to 10.89 s versus teacher-only inference, with 81.33% hybrid accuracy versus 88.00% for the teacher.

Democratizing Fine-Grained Manipulation via Imitation Learning NEU, China
Undergraduate Thesis (Advisor: Prof. Li Zhi) Nov. 2024 – Jun. 2025

- Built and deployed a 6-DOF Koch v1.1 leader-follower system for ACT-based imitation learning, integrating motor calibration, demonstration collection, policy training, and real-time control.
- Optimized task-specific inference pipelines: MobileNetV2 reduced plug-task latency from 20 to 16 ms and supported 60 Hz control, while YOLOv5 segmentation with asynchronous inference achieved 14–15 ms latency for conveyor sorting.
- Evaluated 50 plug-insertion trials and 750 conveyor-sorting operations, achieving 86% plug-grasp success, 66% insertion success, and 89.3% overall sorting success.

Prediction of Ash Content in Coal via Machine Learning NEU, China
Undergraduate Researcher Jan. 2024 – Aug. 2024

- Contributed to a research project on predicting ash content in coal using near-infrared spectral data and PSO-ELM.
- Participated in model development, experimental evaluation, and manuscript preparation; the study achieved $R^2 = 0.98$ and RMSE = 0.79.
- First author of the resulting paper published in Control Engineering of China (Core Journal).

Digital Twin-based Mineral Processing Intelligent Decision System NEU, China
Project Contributor Jan. 2023 – Nov. 2023

- Contributed to ECAS, a correlation-based retrieval module matching equipment states to historical abnormal operating patterns for monitoring and decision support in a mineral-processing digital-twin platform.
- Co-authored ECAS V1.0, registered as Chinese computer software copyright with second-author credit.

PUBLICATIONS

[1] **Y. Gao**, J. Zou, D. Xiao. "Prediction of Coal Ash Composition Based on Near-Infrared Visible Spectroscopy and PSO-ELM." *Control Engineering of China*, 2024. (**Core Journal**)

[2] **Y. Gao**, Z. Feng, K. Mao. Green-Edge-Cloud: Routing without Routers via Confidence Calibration." *ICML 2026 Workshop (EIML)*. **Accepted**.

INTELLECTUAL PROPERTY

[1] **Mineral Processing Equipment Abnormal Pattern Retrieval and Index Correlation Analysis System (ECAS) V1.0**
K. Zhang, **Y. Gao** (2nd Author). *Computer Software Copyright (China)*, Reg. No. 2023SR1469715, Nov. 2023

[2] **Intelligent Venue Resource Allocation and Security Operation Visual Analysis Software (IVVAS) V1.0**
K. Zhang, P. Zhu, **Y. Gao** (3rd Author). *Computer Software Copyright (China)*, Reg. No. 2023SR1421636, Nov. 2023

SELECTED AWARDS

- **Global Merit Award**, Nanyang Technological University 2025-2026
- **National Silver Award (Top 1% nationwide)** Dec. 2023
  China International College Students' Innovation Competition (Team Award)
- **University Academic Scholarship**, Northeastern University 2024-2025
- **University Academic Scholarship**, Northeastern University 2021-2022

TECHNICAL SKILLS

- **Programming & Languages:** Python, C/C++, MATLAB, SQL, LaTeX.
- **AI & LLM Frameworks:** PyTorch, Hugging Face (Transformers, PEFT/LoRA, TRL), vLLM, CUDA, CodeCarbon, Scikit-learn.
- **Robotics & Simulation:** ROS (Robot Operating System), MuJoCo, Gazebo, MobileNet/YOLO.
- **Systems & Deployment:** Linux, Docker, Git, Edge Computing Optimization.
- **Core Algorithms:** LLM Quantization & Knowledge Distillation, Action Chunking Transformer (ACT), Reinforcement Learning, PSO-ELM (Swarm Intelligence), Digital Twin.